

CS 70 离散数学 和 概率论

笔记 12

春季 2026 Course Notes

Counting

The next major topic of the course is probability theory.

Suppose you toss a fair coin a thousand times. How

likely is it that you get exactly 500 heads? And what about 1000 heads?

It turns out that the chances of 500

heads are roughly 2.5%, whereas the chances of 1000 heads are so infinitesimally small that we may as well say that it is impossible. 但是 before you can learn to 计算 或 estimate odds 或 probabilities you must learn to count! That is the subject of this note.

1 Counting Sequences

We start by considering a simple scenario.

We pick $k \leq n$ elements from an n -element set $S = \{1, 2, \dots, n\}$

one at a time while removing the sampled element from S .

This type of sampling is called sampling without

replacement. We wish to count the number of different ways

to do this, taking into account the order in

which the elements are picked. 对于 例子, when $k = 2$, picking 1 和 那么

2 is considered a different

outcome from picking 2 followed by 1.

Another way to ask the question is this: we wish to form an ordered

sequence of k distinct elements, 其中 each 元素 is picked from the 集合

S . How many different such

ordered sequences are there?

If we were dealing cards, the set would be $S = \{1, \dots, 52\}$, 其中

each number represents a card in a deck

of 52 cards. Picking an 元素 of S in this case refers to

dealing one card. 笔记 that once a card is dealt,

it is no longer in the deck and so it cannot be dealt again. So the hand of k

cards that are dealt consists of k

distinct elements from the set S .

对于 the first card, 容易看出 我们有 52 distinct choices. 但是 now the

available choices 对于 the

second card depend upon what card we picked first.

The crucial observation is that regardless of which card

we picked first, 存在 exactly 51 choices 对于 the second card.

所以 the total number of ways of choosing

the first two cards is 52×51 . Reasoning in the same way, 存在

exactly 50 choices 对于 the third card,

regardless of our choices for the first two cards.

It follows that there are exactly $52 \times 51 \times 50$ sequences of

three cards.

In general, the number of sequences of k cards is $52 \times 51 \times \dots \times [52 - (k - 1)]$.

This is an example of the First Rule of Counting:

First Rule of Counting: 如果 an object can be made by a

succession of k choices, 其中 存在 n ways

1

of making the first choice, 和 对于 every way of making the

first choice 存在 n ways of making the

2

second choice, 和 对于 every way of making the first 和 second

choice 存在 n ways of making the

3

third choice, 和 所以 on up to the n -th choice, 那么 the total

number of distinct objects that can be made in

k
 this way is the product $n \times n \times n \times \cdots \times n$.
 $1 \ 2 \ 3 \ k$
 Here is another way of picturing the First Rule of Counting.
 Consider the following tree:
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Level 1 n
 Level 2 n^2
 Level 3 n^3
 Level 4
 It has branching factor n at the 根, n at every 顶点 at the second level, ..., n at every 顶点 at the k -th level.
 Each path from the root to a vertex at level $k+1$ (i.e., a leaf vertex) represents one possible way of making the object by making a succession of k choices. 所以 the number of distinct objects that can be made is equal to the number of leaves in the tree.
 此外, the number of leaves in the tree is the product $n \times n \times n \times \cdots \times n$. For example, if $n=2$, $n=2$, and $n=3$, then there are 12 leaves (i.e., outcomes).
 $1 \ 2 \ 3 \ k \ 1 \ 2 \ 3$

2 Counting Sets

考虑 a slightly different 问题. We would like to pick k distinct elements of $S = \{1, 2, \dots, n\}$ (i.e., without repetition), 但是 we do not care about the order in which we picked the k elements. 对于 例子, picking elements $1, \dots, k$ is considered the same 结果 as picking elements $2, \dots, k$ 和 picking 1 as the last (k -th) element.
 Now how many ways are there to choose these elements to obtain the same outcome?
 When dealing a hand of cards, say a poker hand, it is often more natural to count the number of distinct

hands (i.e., the set of 5 cards dealt in the hand), rather than the order in which they were dealt.
 As we have seen in Section 1, if we are considering order, there are $52 \cdot 51 \cdot 50 \cdot 49 \cdot 48 = 52!$ outcomes. But how many distinct hands of 5 cards are there?
 Here is another way of asking the question: each such 5-card hand is just a subset of S of cardinality 5.
 So we are asking how many 5-element subsets of S are there?

Here is a clever trick for counting the number of distinct subsets of S with exactly 5 elements.
 Create a bin corresponding to each such 5-element subset.
 Now take all these sequences of 5 cards and distribute them into these bins in the natural way:
 each sequence gets placed in the bin corresponding to the set of 5 elements in the 序列. 因此, 如果 the 序列 is $(2, 1, 3, 5, 4)$, 那么 it is placed in the bin labeled $\{1, 2, 3, 4, 5\}$. How many sequences are placed in each bin? The answer is exactly $5!$, since there are exactly $5!$ different ways to order 5 cards.

Recall that our goal was to compute the number of 5-element subsets, which now correspond to the number of bins. We know that there are $52!$ distinct 5-card sequences, and there are $5!$ sequences placed in each bin.

47!

The total number of bins is therefore $52!$.

$47!5!$

This quantity $n!$

is used so often that there is special notation for it:

$(n \choose k)$, pronounced “n choose k,” 和

$\frac{n!}{k!(n-k)!}$

it's known as the 二项的系数. This is the number of ways of

picking k distinct elements from S ,

其中 the order of placement does not matter. Equivalently, it's

the number of ways of choosing k objects

out of a total of n distinct objects, where the order of the choices does not matter.

The trick we used above is actually your Second Rule of Counting:

Second Rule of Counting: 假设 an object is made by a

succession of choices, 和 the order in 其中

the choices are made does not matter. 令 A be the 集合 of ordered

objects 和 令 B be the 集合 of unordered

objects. 如果 存在 an m -to-1 函数 f from A to B , we can count the

number of ordered objects

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(pretending that the order matters) and divide by m (the number of ordered objects per unordered object) to obtain $|B|$, the number of unordered objects.

笔记 that we are assuming that the number of ordered objects

is the same 对于 every unordered object; the

rule cannot be applied otherwise.

Here is another way of picturing the Second Rule of Counting:

$|A|$

$|A| = m |B|$

m

The function f

simply places the ordered outcomes into bins corresponding to the unordered outcomes.

In

our poker hand 例子, f will map $5!$ elements in the 定义域 of the

函数 (the 集合 of ordered 5 card

outcomes) to one 元素 in the 值域 (the 集合 of 5-元素 subsets of S).

The number of elements in the

range of the function is therefore $52!$.

$47!5!$

3 Sampling with Replacement

Sometimes we wish to 考虑 a different scenario of sampling 其中

after we choose an 元素 from

$S = \{1, 2, \dots, n\}$, we throw it back into S so that we can choose it again.

Assume that we are still picking k

elements out of S one at a time and that order matters.

How many distinct sequences of k elements can we

obtain under this new sampling scheme?

We can use the First Rule of Counting. Since we have n choices in

each trial, $n \times n \times \dots \times n = n^k$. Then we have a grand total of n^k

sequences.

$1 \times 2 \times \dots \times k$

This type of sampling is called sampling with replacement; multiple trials can have the same outcome.

Card

dealing is a type of sampling without replacement, since two trials cannot both have the same outcome (one card cannot be dealt twice).

3.1 Coin Tosses

令 us return to coin tosses. How many different outcomes are there 如果 we toss a coin k times? By an 结果, we mean a length- k 序列 consisting of heads 和 tails. 假设 we represent each 结果 by a binary string of length k 其中 0 corresponds to a tail (T) 和 1 a head (H). 对于 例子, the string 001 would represent an 结果 of 2 tails followed by a head. The following 树, in 其中 the leaves are

in 1-to-1 correspondence with the set of possible outcomes, illustrates the experiment for $k=3$:

T H

T H T H

T H T H T H T H

000 001 010 011 100 101 110 111

Here $S=\{0,1\}$. This is a case of sampling with replacement;

multiple coin flips could result in tails (we could pick the element 0 from the set S in multiple trials).

Order also matters; e.g., strings 001 and 010 are

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considered different outcomes.

By the reasoning above (using the First Rule of Counting) we have a total of $n^k = 2^k$ distinct outcomes, since $n=2$.

3.2 Rolling Dice

令' ssay we roll two dice, so $k=2$ and $S=\{1,2,3,4,5,6\}$.

How many possible outcomes are there? In this

setting, ordering matters; obtaining 1 with the first die 和 2 with the second is different from obtaining 2 with the first and 1 with the second.

We are sampling with replacement, since we can obtain the same result on both dice.

The setting is 因此 the same as the coin flipping 例子 above (order matters 和 we are sampling

with replacement), 所以 we can use the First Rule of Counting in the same manner. The number of distinct outcomes is therefore $n^k = 6^2 = 36$.

3.3 Sampling with Replacement, 但是 其中 Order Does 非 Matter

Say you have unlimited quantities of apples, bananas 和 oranges. You want to select 5 pieces of fruit to make a fruit salad. How many ways are there to do this?

In this example, $S=\{1,2,3\}$, where 1 represents

apples, 2 represents bananas, 和 3 represents oranges. $k=5$ 因为 we wish to select 5 pieces of fruit.

Ordering does 非 matter; selecting an apple followed by a banana will lead to the same salad as a banana followed by an apple.

This scenario is trickier to analyze.

It may seem natural to apply the Second Rule of Counting because order does 非 matter. 令' s consider this method. We first pretend that order matters 和 observe the number of ordered objects is 35 as discussed above.

How many ordered options are there for every unordered option?

The problem is that this number differs depending on which unordered object we are considering.

令' ssay

the unordered object is an outcome with 5 bananas.

There is only one such ordered outcome. But if we are

considering 4 bananas and 1 apple, there are 5 such ordered outcomes (represented as 12222, 21222, 22122, 22212, 22221).

Now that we see the Second Rule of Counting will not help, can we look at this problem in a different way?
 Let us first generalize back to our original setting:
 we have a set $S = \{1, 2, \dots, n\}$ and we would like to know

how many ways there are to choose multisets (sets with repetition) of size k .

Remarkably, we can model this

problem in terms of binary strings, as described below.

假设 我们有 one bin 对于 each 元素 of S , 所以 n bins in total. 对于 例子,

如果 we selected 2 apples

和 1 banana, bin 1 would have 2 elements 和 bin 2 would have 1

元素. In order to count the number

of multisets, we need to count how many different ways 存在 to

fill these bins with k elements. We

don't care about the order of the bins themselves, just how many of the k

elements each bin contains. 令 s

represent each of the k elements by a 0 in the binary

string, 和 separations between bins by a 1. The

following picture illustrates an example of placement where $S = \{1, \dots, 5\}$ and $k = 4$:

00110110

Counting the number of multisets is now equivalent to counting the number of placements of the k

0's. We

have just reduced what seemed like a very complex problem to a

问题 about a binary string, simply by

looking at it from a different perspective!

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How many ways can we choose these locations? The length of

our binary string is $k + n - 1$, 和 we are

choosing which k locations should contain 0's.

The remaining $n - 1$ locations will contain 1's. Once we pick

a location 对于 one zero, we cannot pick it again; repetition

is 非 allowed. Picking location 1 followed by

location 2 is the same as picking location 2 followed by

location 1, 所以 ordering does 非 matter. It follows

that all we wish to 计算 is the number of ways of picking k

elements from $k + n - 1$ elements, without

replacement 和 其中 the order of placement does 非 matter. This

is 给定 by

$\binom{k+n-1}{k}$

, as discussed in

Section 2.

This is therefore the number of ways in which we can choose multisets of size k from set S .

Returning to our example above, the number of ways of picking 5 pieces of fruit is exactly

$\binom{3+5-1}{3} = \binom{7}{3}$

$=$

$\binom{7}{4} = 35$

.

5 5

Notice that we started with a problem which seemed very different from previous examples, 但是, by viewing

it from a certain perspective, we were able to use previous techniques (those used in counting sets) to find a 解答!

In a sense, we have found the zeroth rule of counting:

Zeroth Rule of Counting:

If a set A can be placed into a one-to-one correspondence with a set B (i.e., you

can 求 a bijection between the two—an invertible pair of maps that map elements of A to elements of B and vice-versa), 那么 $|A|=|B|$.

This is the very heart of what it means to count 和 is key to many combinatorial arguments, as we will explore further in the next section.

4 Combinatorial Proofs

Combinatorial proofs are interesting because they rely on intuitive counting arguments rather than tedious algebraic manipulation.

They often feel like proofs by stories—the same story, told from multiple points of view.

As a warm-up, 令's prove the Binomial Theorem using counting arguments:

Theorem 12.1 (Binomial Theorem). For all $n \in \mathbb{N}$,

$$(a+b)^n = \sum_{k=0}^n \binom{n}{k} a^k b^{n-k}.$$

$$(a+b)^n = \sum_{k=0}^n \binom{n}{k} a^k b^{n-k}.$$

k

$$k=0$$

证明.

The left-hand side can be written as the n -fold product $(a+b)(a+b) \cdots (a+b)$.

Label these n factors

as $f_1, f_2, \dots, f_n$. When $(a+b)(a+b) \cdots (a+b)$ is expanded out via multiplication, it will result in a sum

$$\sum_{i_1, i_2, \dots, i_n} a^{i_1} b^{i_2} \cdots a^{i_n} b^{i_n}$$

over 2^n monomials, 和 to each monomial in the sum, each factor f_i contributes either an a 或 a b . 对于

i

例子, in $(a+b)(a+b) = a^2 + ab + ba + b^2$, a^2 is formed by multiplying a from f_1 and a from f_2 ; ab is

$$a^2$$

formed by multiplying a from f_1 and b from f_2

; ba is formed by multiplying b from f_1 and a from f_2 ; b^2 is

$$a^2$$

formed by multiplying b from f_1 and b from f_2 .

In general, the monomial $a^k b^{n-k}$ is obtained when k copies

$$a^k b^{n-k}$$

of a 's and $n-k$ copies of b 's are multiplied. There are exactly

$$\binom{n}{k}$$

ways to choose k copies of a 's and $n-k$

k

copies of b 's from the factors $f_1, f_2, \dots, f_n$.

$$a^k b^{n-k}$$

This well-known result has numerous applications. 对于 例子, the following combinatorial identity can

be easily derived using Theorem 12.1:

Corollary 12.1. For all $n \in \mathbb{N}$,

$$\sum_{k=0}^n \binom{n}{k} = 2^n.$$

$$\sum_{k=0}^n \binom{n}{k} = 2^n.$$

k

$$k=0$$

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证明. Set $a=1$ and $b=1$ in Theorem 12.1.

In what follows, we will 考虑 more sophisticated versions of combinatorial proofs. The key idea is to

证明 that the two sides of a 给定 方程 correspond to two different ways to count the same kind of objects.

In more advanced settings, the goal is to establish a bijection between the set of objects counted on the left-hand side of an identity with the 集合 of objects counted on the right-hand side. We illustrate these

concepts below.

4.1 Combinatorial Identities

Using combinatorial arguments, we can prove complex facts, such as

(cid:18) (cid:19) (cid:18) (cid:19) (cid:18) (cid:19)

(cid:18) (cid:19)

$n \choose n-1$ $n \choose 2$ k

$= 1 + \dots + n$, (1)

$k+1 \choose k$

which is known as the hockey-stick identity.

You can directly verify this identity by algebraic manipulation.

但是 you can also do this by interpreting what each side means

as a combinatorial process. The left-hand

side of (1) is just the number of ways of choosing a subset X

with $k+1$ elements from a set $S = \{1, \dots, n\}$.

Let us think about a different process that results in the choice of a subset with $k+1$ elements.

We start by

picking the lowest-numbered element of X .

If we picked 1, to finish constructing X , we must now choose k

elements out of the remaining $n-1$ elements of S greater than 1; this can be done in

(cid:0) $n-1$ (cid:1)

ways. If instead

k

the lowest-numbered element we picked is 2, 那么 我们有 to choose k

elements from the remaining $n-2$

elements of S greater than 2, which can be done in

(cid:0) $n-2$ (cid:1)

ways. Moreover all these subsets are distinct from

k

those 其中 the lowest-numbered element was 1. 所以 we should add the

number of ways of choosing each

to the grand total. Proceeding in this way, we split up the

process into cases according to the first (i.e.,

lowest-numbered) object we select, to obtain:

element 1, leading to

(cid:0) $n-1$ (cid:1)

remaining ways,

k

element 2, leading to

(cid:0) $n-2$ (cid:1)

remaining ways,

k

First element selected is element 3, leading to (cid:0) $n-3$ (cid:1)

remaining ways,

k

.

元素 $(n-k)$, leading to

(cid:0)k(cid:1)

remaining ways.

k

These factors correspond to the terms on the right-hand side of (1). (笔记 that the lowest-numbered object

we select cannot be higher than n - k as we have to select k+1 distinct objects.)

Another combinatorial identity we will prove is

(cid:18) (cid:19) (cid:18) (cid:19) (cid:18) (cid:19)

n n n

+ + • • • + = 2ⁿ. (2)

0 1 n

This identity actually follows immediately from setting a=1

and b=1 in the 二项的 定理 above,

但是 we will provide a more direct combinatorial 证明. 假设 我们有 a 集合 S with n distinct elements.

On the left-hand side of (2),

(cid:0)n(cid:1)

counts the number of ways of choosing a 子集 of S of size exactly i; 所以

i

the sum on the left-hand side counts the total number of subsets (of any size) of S.

We claim that the right-hand side of (2) does indeed also count the total number of subsets.

To see this, just

identify a subset with an n-bit vector, where i in each position

j we put a 1 if the j-th element is in the subset,

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Table 1: Permutations of {1, 2, 3} and the number of fixed points.

(π , π , π) #fixed points

1 2 3

(1, 2, 3) 3

(1, 3, 2) 1

(2, 1, 3) 1

(2, 3, 1) 0

(3, 1, 2) 0

(3, 2, 1) 1

and a 0 otherwise.

So the number of subsets is equal to the number of n-bit vectors, which is 2ⁿ

(there are 2

options for each bit).

令 us look at an 例子, 其中 S = {1, 2, 3} (所以 n = 3). Enumerate all 2³ = 8 possible subsets of S:

{}, {1}, {2}, {3}, {1, 2}, {1, 3}, {2, 3}, {1, 2, 3}. The eight 3-bit vectors corresponding to these subsets are

(0, 0, 0), (1, 0, 0), (0, 1, 0), (0, 0, 1), (1, 1, 0), (1, 0, 1), (0, 1, 1), (1, 1, 1), respectively. On the left-hand side of

(2), the term

(cid:0)3(cid:1)

counts the number of ways to choose a subset of S with 0 elements; there is only one such

0

子集, namely the empty set = {}. There are (cid:0)3(cid:1)

= 3 ways of choosing a subset with 1 element, (cid:0)3(cid:1) = 3

1 2

ways of choosing a subset with 2 elements, 和

(cid:0)3(cid:1)

=1 way of choosing a subset with 3 elements (即,
 3
the subset consisting of every element of S).
Summing, we get $1+3+3+1=8$, as expected.

4.2 Permutations 和 Derangements

问题:

Suppose we collect the homeworks of n students, randomly shuffle them, and return them to the students. 如果 we repeated this experiment many times, 那么 on average how many students would receive their own homework?

At first this may seem like a rather challenging 题目, 但是 you will be able to 答案 this 问题 in a

few lines later in the course once you have some basic tools of probability theory under your belt.
For now,

令' s translate the question into a mathematical statement and consider a related counting problem.
Label the homeworks as $1, 2, \dots, n$ 和 令 π 表示 the 作业 即 returned to the i -th student.

i
那么, the vector $\pi = (\pi_1, \dots, \pi_n)$ corresponds to a permutation of the set $\{1, \dots, n\}$. Note that $\pi_1, \dots, \pi_n \in \{1, 2, \dots, n\}$.

$\{1, 2, \dots, n\}$ are all distinct, 所以 each 元素 in $\{1, \dots, n\}$ appears exactly once in the permutation π . In total, there are $n!$ distinct permutations of $\{1, \dots, n\}$.

In this setting, the i -th student receives their own 作业 如果 和 only 如果 $\pi_i = i$. 那么 the 问题 “How

i
many students receive their own homework?” translates into the question of how many indices (i ' s) satisfy $\pi_i = i$. These are known as fixed points of the permutation π . To illustrate the idea concretely, let us consider

i
the 例子 $n=3$. 表 1 shows a 完全的 list of all $3!=6$ permutations of $\{1, 2, 3\}$, together with the corresponding number of fixed points.

We now consider an interesting counting problem concerning derangements, which are defined as follows.
Definition 12.1 (Derangement).

A permutation with no fixed points is called a derangement.

Let D_n denote the number of derangements of $\{1, 2, \dots, n\}$.

As can be seen in Table 1, $D_2 = 1$; specifically,

$n=3$
the derangements are $(2, 3, 1)$ 和 $(3, 1, 2)$. Our goal is to 求 a formula 对于 D_n as a 函数 of n 对所有

$n \in \mathbb{Z}^+$. We will learn two different ways to solve this problem.
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The first method involves a combinatorial proof.

For concreteness, 令' s consider the case of $n=4$. Table 2a lists all derangements of $\{1, 2, 3, 4\}$, with each row corresponding to a derangement. In the first derangement, 笔记 that the entries shown in blue corresponds to a derangement of $\{1, 2\}$. In the second 和 third derangements, the entries shown in red are in 1-to-1 correspondence with the derangements of $\{1, 2, 3\}$, shown in Table 2b.

Note that the red 4 in Table 2a satisfies the same constraint as the number 3 in Table 2b;

specifically, $\pi = 4$ in the former and $\pi = 3$ in the latter. In summary, the above 讨论 suggests that

3 3

D is related to D and D .

The following theorem formalizes this relationship.

4 2 3

Table 2: List of derangements. (a) $n=4$. (b) $n=3$.

(a) (b)

$\pi \quad \pi \quad \pi \quad \pi \quad \pi \quad \pi \quad \pi$

1 2 3 4 1 2 3

2 1 4 3 2 3 1

2 4 1 3 3 1 2

4 1 2 3

4 3 2 1

3 4 2 1

2 3 4 1

3 4 1 2

4 3 1 2

3 1 4 2

Theorem 12.2. For an arbitrary positive integer $n \geq 3$, the number D of derangements of $\{1, \dots, n\}$ satisfies

n

$D = (n-1)(D + D)$. (3)

$n \quad n-1 \quad n-2$

证明.

We provide a combinatorial proof by establishing relevant bijections.

In a derangement $(\pi_1, \dots, \pi_n)$

$1 \leq n$

of $\{1, \dots, n\}$, 假设 $\pi_j = j \in \{1, \dots, n-1\}$; i.e., j is what n maps to under the permutation. 存在

n

$n-1$ choices 对于 j , 和 this gives rise to the multiplicative factor $(n-1)$ in (3). (The case of $j = n$ is

excluded, 因为 a derangement cannot have $\pi_n = n$.) 对于 a 给定 value of $j \in \{1, \dots, n-1\}$, 存在 two

n

cases to consider:

Case 1: 如果 $\pi_n = n$, 那么 j 和 n get swapped, 和 存在 a 1-to-1 correspondence between the 集合 of

j

derangements of the remaining elements and the set of derangements of $\{1, \dots, n-2\}$.

This gives rise to

D .

$n-2$

Case 2: 如果 $\pi_n = n$, 那么 n satisfies the same constraint as j , 所以 存在 a 1-to-1 correspondence between

j

the set of derangements of $\{1, \dots, j-1, j+1, \dots, n\}$ and the set of derangements of $\{1, \dots, n-1\}$.

This

gives rise to D .

$n-1$

This completes our derivation of (3).

Together with the boundary conditions $D=0$ and $D=1$ (check this), D can be determined efficiently (in

$1 \leq n$

linear time in n) using the recursion in (3) and memoization.

In fact, the recursion can be solved analytically

toyield(seeExercise2)anexplicitformula:

$$n \choose k$$

$$D = n! \sum_{k=0}^n \frac{(-1)^k}{k!} \binom{n}{k}$$

$$n$$

$$k!$$

$$k=0$$

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Comingupwiththerecursionin(3)andsolvingitrequiresomecreativity.

Asmentionedearlier, thereisan

alternatewaytotacklethiscountingproblem, whichturnsouttobeamorestraightforwardapproach.

This

alternatemethodutilizesapowerfultoolcalledtheprincipleofinclusion-exclusion, whichwenowdescribe.

5 The Principle of Inclusion-Exclusion

Inthissectionwewilllearnausefulformulathatcanbeemployedtocountobjectswithoutover-counting.

We startwithafamiliarexample. Let A and A

betwosubsetsofthesamefiniteset A , andsupposewewant

1 2

to count the number of elements in $A \cup A$. 如果 A 和 A are

disjoint, 那么 clearly $|A \cup A| = |A| + |A|$.

1 2 1 2 1 2 1 2

If A and A

havecommonelements, 然而, thenthoseelementswouldbecountedt看 twiceintheprevious

1 2

formula. Tocorrectforthisovercounting, weneedtosubtract $|A \cap A|$

from $|A| + |A|$, thusyielding

1 2 1 2

$$|A \cup A| = |A| + |A| - |A \cap A|$$

1 2 1 2 1 2

Theprincipleofinclusion-exclusiongeneralizesthisconcepttoanarbitrarynumberofsubsets.

Theorem12.3(Inclusion-Exclusion). Let $A_1, \dots, A_n$

be arbitrarysubsetsofthesamefiniteset A . 那么,

1 n

n

$$|A_1 \cup \dots \cup A_n| = \sum_{i=1}^n |A_i| - \sum_{1 \leq i < j \leq n} |A_i \cap A_j| + \sum_{1 \leq i < j < k \leq n} |A_i \cap A_j \cap A_k| - \dots + (-1)^{n+1} |A_1 \cap \dots \cap A_n|$$

1 n i ∈ S i

k=1 S = {1, ..., n}: |S|=k

Remarks:

1. The inner summation in (5) is over all size- k subsets of

$\{1, 2, \dots, n\}$. More explicitly, (5) can be

writtenas

n

$$|A_1 \cup \dots \cup A_n| = \sum_{i=1}^n |A_i| - \sum_{1 \leq i < j \leq n} |A_i \cap A_j| + \sum_{1 \leq i < j < k \leq n} |A_i \cap A_j \cap A_k| - \dots + (-1)^{n+1} |A_1 \cap \dots \cap A_n|$$

+ . . .

1 n i i j j k i j k l

i=1 i < j i < j < k i < j < k < l

+ (-1)^{n+1} |A_1 \cap A_2 \cap \dots \cap A_n|, (6)

1 2 n

其中 $\sum$ denotessummingoverall $i, j \in \{1, \dots, n\}$ suchthat $i < j$, andsoon.

i < j

2. ThestandardwaytoproveTheorem12.3isviainductiononn.

Here, wewillprovideacombinatorial

证明.

证明. First, 笔记 that 如果 $a \in A_1 \cup \dots \cup A_n$, 那么 $a \in A_i$ 对所有

$i \in \{1, \dots, n\}$ 和 因此 $a \in A_1 \cap A_2 \cap \dots \cap A_n$ 对于 any

$1 \leq i \leq n$
 $S = \{1, \dots, n\}$, so it is not counted by the right-hand side (RHS) of (5).
 We now show that if $a \in A_1 \cup \dots \cup A_n$,
 $1 \leq i \leq n$
 then it is counted exactly once by the RHS. Define the index set $M = \{1, \dots, n\}$ as
 $M = \{i \in \{1, \dots, n\} \mid a \in A_i\}$;
 i
 that is, $a \in A_i$ if and only if $i \in M$. Note that $m = |M| \geq 1$, since $a \in A_1 \cup \dots \cup A_n$. Clearly, if $S = M$, 那么
 $1 \leq i \leq n$
 $a \in \bigcap_{i \in S} A_i$ (i.e., 如果 S contains an index i 非 in M , 那么 $a \notin \bigcap_{i \in S} A_i$, 所以
 $a \in \bigcap_{i \in S} A_i$). 此外, $a \in \bigcap_{i \in S} A_i$ 对于
 $i \in S$ 且 $i \in S$ 且 $i \in S$ 且
 any non-empty subset $S \subseteq M$.
 因此, the number of times that a is counted on the RHS of (5) is
 $\sum_{k=1}^m \binom{m}{k} (-1)^{k-1} = \sum_{k=1}^m \binom{m}{k} = 2^m - 1$,
 k
 $k=1 \leq S \subseteq M: |S|=k$

where the first equality follows from the fact that the number of distinct k -element subsets of M is
 $\binom{m}{k}$,
 , 当...时
 k
 these two equalities follow from Corollary 12.1.
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Application: Derangements Revisited

We now illustrate the power of 定理 12.3 by applying it to
 derive the formula shown in (4) 对于 the
 number D_n of derangements of $\{1, \dots, n\}$.

n
 令 A_i 表示 the 集合 of all permutations of $\{1, \dots, n\}$ with i being
 a fixed point. 那么, 因为 存在 $n!$
 i
 distinct permutations of $\{1, \dots, n\}$ 和 $|A_1 \cup A_2 \cup \dots \cup A_n|$ counts
 the number of permutations with at least
 $1 \leq i \leq n$
 one fixed point, we have
 $D_n = n! - |A_1 \cup A_2 \cup \dots \cup A_n|$. (7)
 $n \geq 1$
 What is the cardinality of A ?
 We know that i maps to i , while the remaining $n-1$ elements can be arranged
 i
 in an arbitrary way. 所以, $|A_i| = (n-1)!$. Similarly, for $i =$
 j , i maps to i and j maps to j , while the remaining
 i
 $n-2$ elements can be arranged in an arbitrary way, 所以 我们得出结论
 $|A_i \cap A_j| = (n-2)!$. In general, 对于
 $i \in S$
 any 子集 $S \subseteq \{1, \dots, n\}$ of size $|S|=k$, $| \bigcap_{i \in S} A_i | = (n-k)!$, 当...时 存在
 $(cid:0)n \ (cid:1)$
 such subsets. 因此, the
 $i \in S$ 且 $|S|=k$
 inclusion-exclusion formula from (5) gives
 $|A_1 \cup A_2 \cup \dots \cup A_n| = \sum_{k=1}^n \binom{n}{k} (-1)^{k-1} (n-k)! = n! \sum_{k=1}^n \frac{(-1)^{k-1}}{k!}$.
 $1 \leq k \leq n$
 $k! (n-k)! k!$

$k=1 \quad k=1 \quad k=1$

Substituting this result into (7) gives

$n \binom{n-1}{k} k! = n \binom{n-1}{k}$

$D^n = n! \sum_{k=0}^n \frac{1}{k!} D^k$.

n

$k! \quad k!$

$k=1 \quad k=0$

6 Exercises

1. Provide a direct combinatorial proof of Corollary 12.1.

2. Using (3) and the boundary conditions $D = 0$ and $D = 1$, derive

(4). [提示: 对于 $n \geq 2$, 定义

$1 \leq 2$

$Q_n = D^n$ and obtain a recursion for Q_n .]

$n \leq n-1 \leq n$

3. Prove Theorem 12.3 using induction.

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